The only other available operation is iterating through bags element: that does not change the container, but the order in which elements are returned is not guaranteed: it could be insertion order, but it could also be completely random each time the bag is iterated upon.

So, a *bag* can be described in terms of priority queues: one for which element removal is disabled and priority is given by a random number from a uniform distribution.

D.2 Stack

A stack is a collection that returns its elements according to the LIFO (Last In First Out) policy. This means that the elements in the stack will be returned in the reverse of the order they were added to it. This makes stacks pretty useful when you need to reverse a sequence.

We can see a stack as a specialized *PQ* where elements' priority is given by the order in which they are inserted (most recently inserted elements have highest priority).

When comparing a stack to a bag the first difference is, in fact, that for stacks order matters.

The second one is that, while it is possible to iterate through the whole stack at once, it's also possible to remove elements from a stack. In particular, a stack is a limited access data structures, because it's only possible to add or remove elements to/from the top of the stack.